

- George:** Wow, how many of these tips have you gotten?
- Jim:** My broker gave me three great stock picks over the past month or so. Each stock is up now, by over 10 percent.
- George:** That's a great record. My broker seems to give me one bad pick for every good one. It sounds like I need to talk to your broker; she has a much better record!

Analysis: (As per Case Study 1.) As we'll see in a moment, this conversation exemplifies sample-size neglect representativeness bias. Jim's description has prompted George to arrive at a faulty judgment regarding the success rate of Jim's broker/analyst. George is impressed, but his assessment is based on a very small sample size; the recent, successful picks Jim cites are inevitably only part of the story. George concluded that Jim's broker is successful because Jim's account of the broker's and analyst's performances seems *representative* of the record of a successful team. However, George disproportionately weighs Jim's testimony, and if he were to ask more questions, he might discover that his conclusion draws on too small a sample size. In reality, the analyst that Jim is relying on happens to be one who covers an industry that is popular at the moment, and *every* stock that this analyst covers has enjoyed recent success. Additionally, Jim neglected to mention that last year, this same broker/analyst team made a string of three *losing* recommendations. Therefore, both Jim's and George's brokers are batting 50 percent. George's reasoning demonstrates the pitfalls of sample-size neglect representativeness bias.

Implications for Investors. Both types of representativeness bias can lead to substantial investment mistakes. In the following section, we list examples of behaviors, attributable to base-rate neglect and sample-size neglect, respectively, that can cause harm to an investor's portfolio. Advice on these four areas will come later.

RESEARCH REVIEW

In *Judgment under Uncertainty: Heuristics and Biases*, Daniel Kahneman, Paul Slovic, and Amos Tversky apply representativeness bias to the world of sports. The concepts brought forward in this book also translate easily to finance.